HCSA test pieces, Rev F, 8 November 2026

The T1 seam slices (75, 150, 225 mm), the 550 mm seal article with its grip legs and dummy hardpoint, and the pull blocks, all at the 180 mm frame.
Drawn from hcsa_params_revF.json (5 Nov). What changed from the 24 Oct Rev E draft: frame depth 140 to 180, gland 7.1 × 4.0, seam bolt NAS6705U27 at 4 kN,
the pull blocks grow to 160 tall to span the legs, run 2 pulls through pinned lugs (RE-03 rev B), the 550 article has legs and a dummy hardpoint (2 + 2 bolts).

Sheets

TP-00	Cover, notes, tolerances, parts list
TP-01	T1 frame B (glands): section, joint face, legs; 75 / 150 / 225 table
TP-02	T1 frame A and the slice as assembled, pull lines, screws
TP-03	550 mm seal article with legs and the dummy hardpoint, both fitting depths
TP-04	Pull blocks (rev B interface) and the lug, fork and pin F-01 to F-03
STEP	out/step: one file per part; out/asm: one assembly per article

General notes

1. Material: frames 6061-T651 plate (as drawn, keeps the FEA twins valid). Pull blocks 4140 quenched and tempered 28-32 HRC (as drawn).
   DFM option for Marco: blocks in 6061-T651 (one material, 65 % lighter: 19 kg instead of 55 kg each on the 550 article).
2. Machining order for the frames: rough, rest 24 h, finish; the joint face is the last cut (fly-cut), so the part is flat after the stress is out.
3. Inside corners web-to-flange and web-to-lip R6. Every hole 1 × 45 deg at both ends. All other edges 0.5 mm break. Gland edges deburred only, no break.
4. Surface finish: joint face and gland faces Ra 0.8 um. Elsewhere Ra 3.2.
5. Hardware (test articles, McMaster-Carr or equal): ISO 4762 M12 × 35 class 12.9 socket screws + ISO 7089 hardened washers into the blocks;
   M12 × 45 class 12.9 for the lug plates; seam bolts NAS6705U27 with KFN670 nuts and two hardened washers (flight part, A286).
6. Seals: Parker BA500-70 butyl cord 0.210 in, two per article, RTV566 bead with SS4155 primer. End dams on the slices (M8 rod, fabrication plan 2.3).
7. Quantities: one of each slice (75, 150, 225) and one 550 article: 8 frames, 8 blocks. Fixture parts F-01 to F-03 are shared: 2 lugs, 2 forks, 2 pins + 1 spare each.

Tolerances: tight only where the seal or the fit needs it (proposal for Marco, 17 Nov)

Feature	As drawn 24 Oct	Proposed	Why
Joint face flatness	0.05 over the length	0.10 per 150 mm, 0.15 over 550	0.1 mm of gap moves the squeeze 2 points of 25 %
Joint face finish	Ra 0.8	Ra 0.8 (keep)	Parker face-seal finish
Gland depth 4.0	+-0.05	+-0.05 (keep)	sets the 25 % squeeze; a shallow groove, easy to hold
Gland width 7.1	+-0.05	+-0.10	fill 78.7 % moves 1 point per 0.1 mm
Gland position z 18, 32	+-0.05	+-0.15	the cords only have to meet the flat face of frame A
Seam bolt holes	8.4 reamed, pos +-0.2 (old)	8.5 drilled, pos +-0.2 (as modelled)	a clamp bolt, not a shear pin, in the test articles
Leg holes	13.5 (old)	14.0 drilled, pattern +-0.2 (as modelled)	M12 screws need play to find the block threads
Hardpoint bores (550)	40 H7	40 +0.10 / 0	the dummy pins are fitted loose by design
Block bearing faces	parallel 0.05	parallel 0.10, height 160 +-0.1	the legs must not be sprung when the screws go in
Everything else	ISO 2768-f	ISO 2768-m	legs, lengths, block outline: nothing seals on them

Temperature: all sizes at 20 C. 7 C of difference moves 550 mm of aluminium by 0.09 mm (23.6e-6 per K), so the long parts are measured at 20 C (Marco, 7 Nov).

Parts list (masses from the solids)

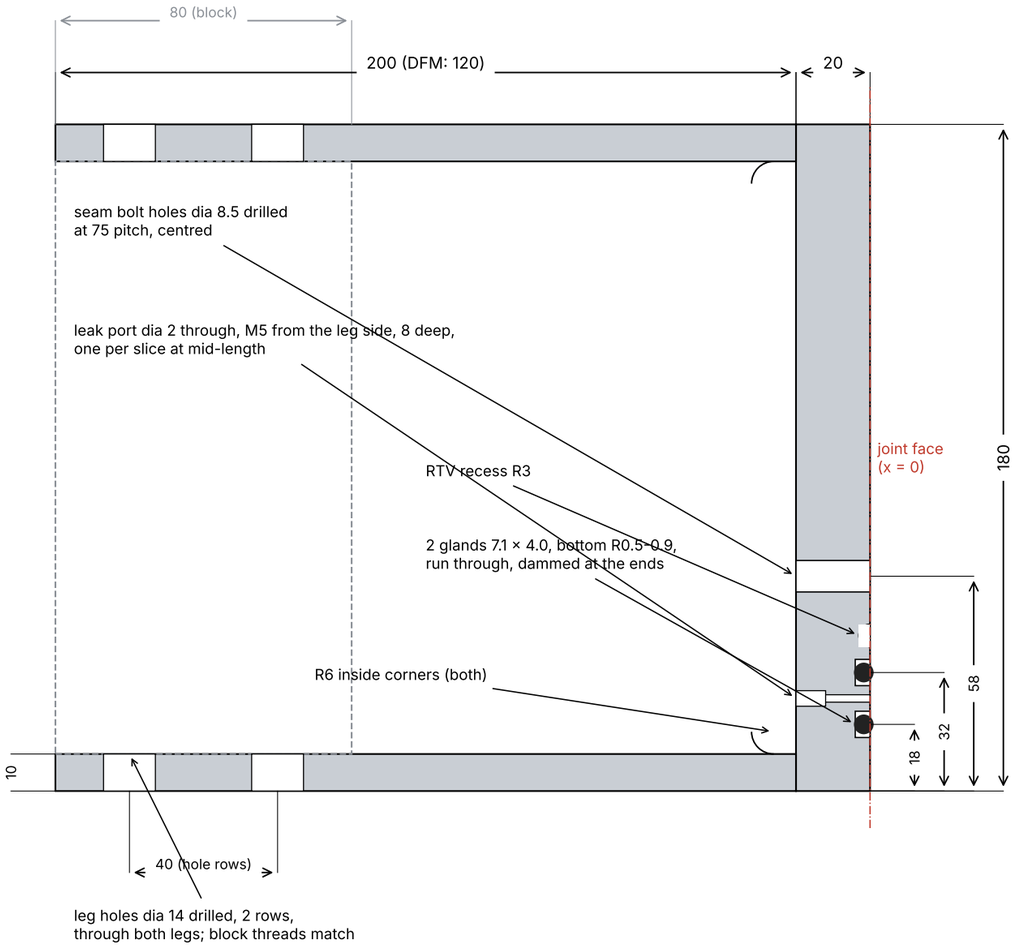
Part	Description	Material	Qty	Mass each kg	Stock blank (mm)
T1-075-FB	Frame B (glands, RTV recess, leak port), 75 mm, 180 deep, legs 200	6061-T651	1	1.49	190 × 230 × 81
T1-075-FA	Frame A (flat joint face), 75 mm, 180 deep, legs 200	6061-T651	1	1.51	190 × 230 × 81
T1-075-PB	Pull block 160 × 80 × 75, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	7.22	165 × 85 × 81
T1-150-FB	Frame B (glands, RTV recess, leak port), 150 mm, 180 deep, legs 200	6061-T651	1	3.02	190 × 230 × 156
T1-150-FA	Frame A (flat joint face), 150 mm, 180 deep, legs 200	6061-T651	1	3.04	190 × 230 × 156
T1-150-PB	Pull block 160 × 80 × 150, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	14.75	165 × 85 × 156
T1-225-FB	Frame B (glands, RTV recess, leak port), 225 mm, 180 deep, legs 200	6061-T651	1	4.52	190 × 230 × 231
T1-225-FA	Frame A (flat joint face), 225 mm, 180 deep, legs 200	6061-T651	1	4.57	190 × 230 × 231
T1-225-PB	Pull block 160 × 80 × 225, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	22.21	165 × 85 × 231
SA-550-FB	Frame B (glands, RTV recess, leak port), 550 mm, 180 deep, legs 200	6061-T651	1	10.87	190 × 230 × 556
SA-550-FA	Frame A (flat joint face), 550 mm, 180 deep, legs 200	6061-T651	1	10.98	190 × 230 × 556
SA-550-PB	Pull block 160 × 80 × 550, M12 from both bearing faces, M24 run 1, lug rows run 2	4140 QT	2	54.54	165 × 85 × 556
F-01	Lug plate 80 × 100 × 20 (RE-03 rev B)	4140 QT	2	1.05	from the parts library STEP
F-02	Fork with M24 × 3 for the Instron adapter	4140 QT	2	1.80	parts library STEP
F-03	Pin 20 h6 × 70 with shoulder and E-clip groove	17-4PH H900 or 4140 QT	2	0.20	parts library STEP
SA-550-HPmoved-*	The same three 550 parts with the dummy fitting bores at z 76 instead of 140 (make these: moved fitting agreed 10 Nov)	as above	-	same	same

HCSA TP-00 Cover: sheet list, notes, tolerances, parts list

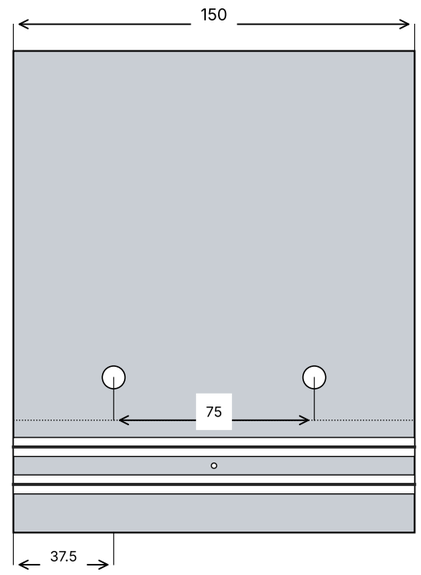
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: for quotation; tolerances and DFM items are proposals for Marco, 17 Nov	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

Section through a bolt station, looking along the seam (1:1 on A3)

space side (z 0) at the bottom, cabin side (z 180) at the top; the seat lip (z 170-180) doubles as the second grip leg



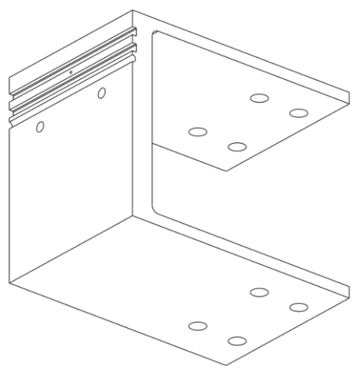
Joint face of the 150 slice (frame B)



Leg holes (both legs, through), measured from the slice end along the seam and from the joint face across the leg

Slice	Seam bolts (from the end)	Leg holes, inner row x 140	Leg holes, outer row x 180	Threads / face
75	37.5	18.75, 56.25	16.5, 58.5 (OPEN-6, moved for the M24 wall)	4
150	37.5, 112.5	37.5, 112.5	37.5, 112.5	4
225	37.5, 112.5, 187.5	37.5, 112.5, 187.5	37.5, 112.5, 187.5	6
550 (TP-03)	50, 125, 425, 500	7 at 75 pitch, 50 to 500	same	14

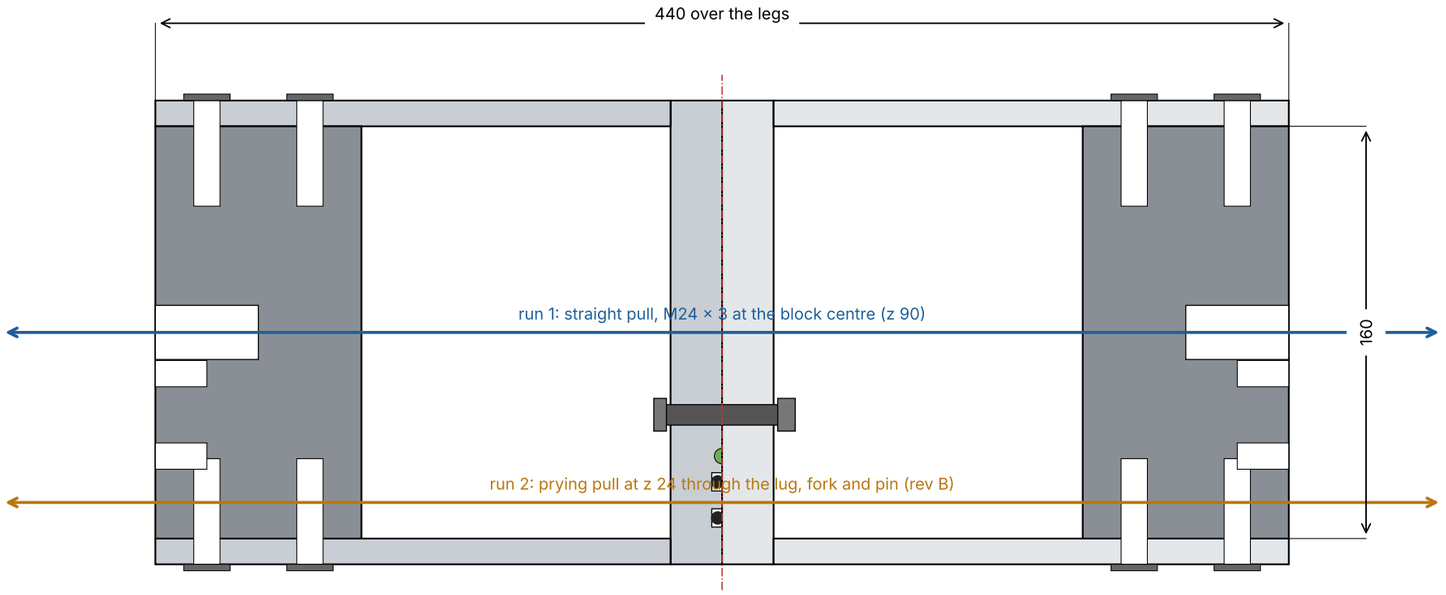
T1-150-FB, from the STEP (legs toward the viewer, joint face at the back)



Frame A (TP-02) is the mirror image without the glands, the recess and the port.
Stock 6061-T651 plate 190 × 230 x (L + 6); with 120 mm legs, 190 × 150 x (L + 6).
The slice is flat, so z equals the joint-plane coordinate z' of the flight frame: the test measures the same joint face the station has.

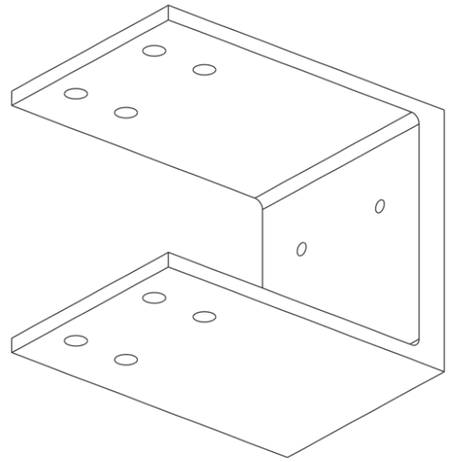
HCSA TP-01 T1 frame B (glands), 180 deep: section, joint face, legs	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: geometry from build_testpieces.py; 75 / 150 / 225 as a table	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

Section through a bolt station: frame B (left, glands), frame A (right), the two pull blocks and the seam bolt



M12 × 35 class 12.9 screws with hardened washers through the 14 mm leg holes into the block threads (4 to 14 per face).
The screws carry the pull in shear; the washers keep the head pressure on the 6061 leg under 132 MPa (RE-01-2 check).

T1-150-FA (flat face), from the STEP

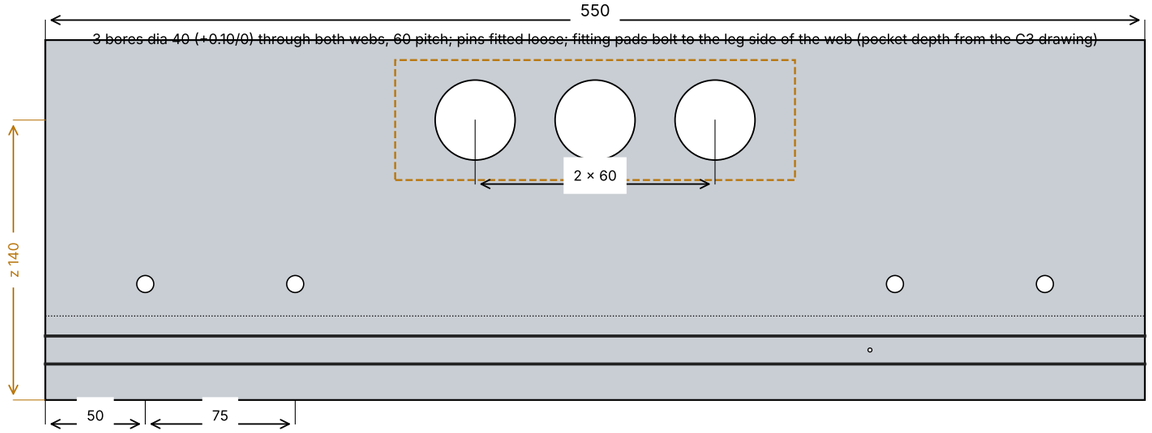


Pull block, 160 × 80 × 150 (TP-04)

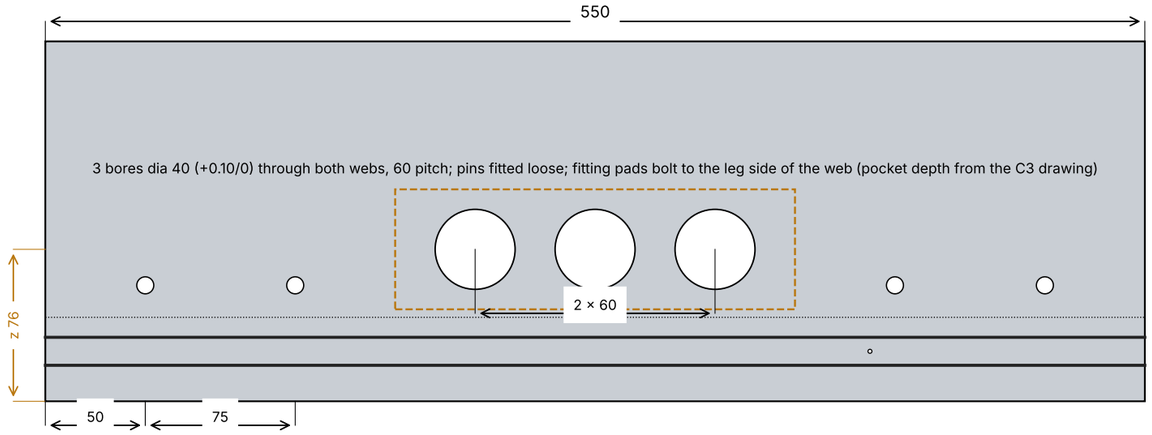
Why the block grew. On the 140 frame the block spanned the flange and the lip, 83 mm apart. On the 180 frame the lip is at the cabin face (z 170-180, P-06), so the span is 160 mm. The single M24 for run 1 sits at the block centre, which puts the straight pull at z 90; the D1 and HP1 twins pull at the frame middle (76 mm), so run 1's pull line is 14 mm off the twin's. Run 2 (the one the seal design turns on) is unchanged at z 24.
The 2.0 and 3.5 mm walls round the two M24 threads of the 24 Oct blocks are gone: rev B has one M24 and the z 24 pull goes through the lugs.
DFM questions for Marco: block in 6061-T651 instead of 4140 (19 kg instead of 55 kg for each 550 block; the M12 and M24 threads in aluminium carry the 100 kN frame load with margin); legs 120 instead of 200 (stock 190 × 150 instead of 190 × 230; the block moves in; the twins' fixture legs change).

HCSA TP-02 T1 frame A and the slice as assembled, pull lines, screws	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: block 160 tall; run 1 M24 at z 90; run 2 through the lugs at z 24 (rev B)	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

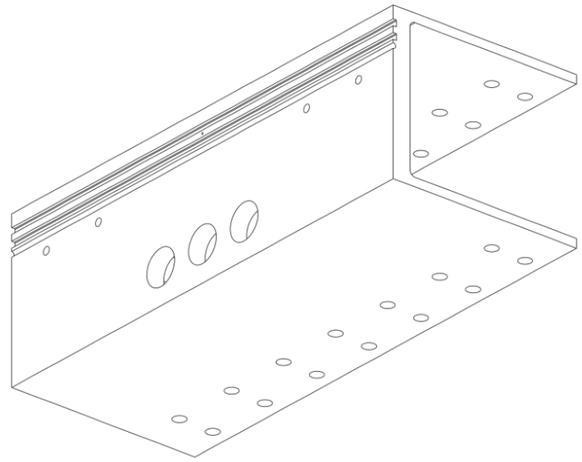
Joint face of frame B, dummy hardpoint as drawn (RE-06-1: fitting 110 to 170 deep)



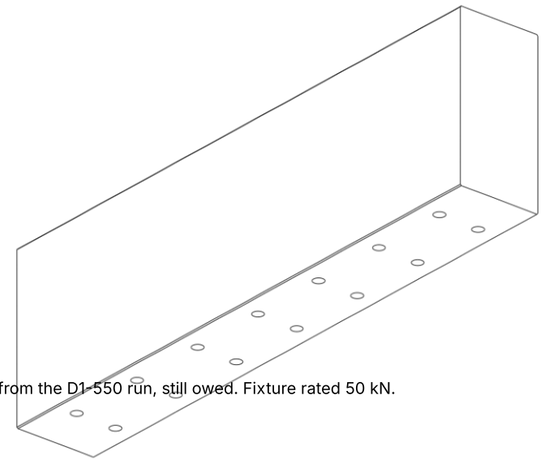
Joint face of frame B, dummy hardpoint moved onto the pull line (HP1M study: 46 to 106 deep)



SA-550-FB from the STEP: legs, 14 + 14 leg holes, 4 seam bolts, 3 pin bores



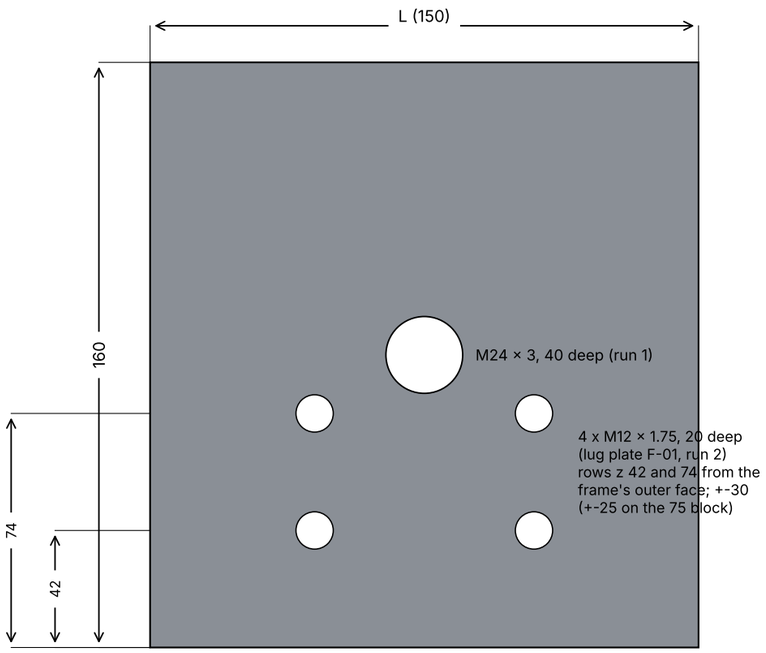
SA-550-PB, 160 × 80 × 550



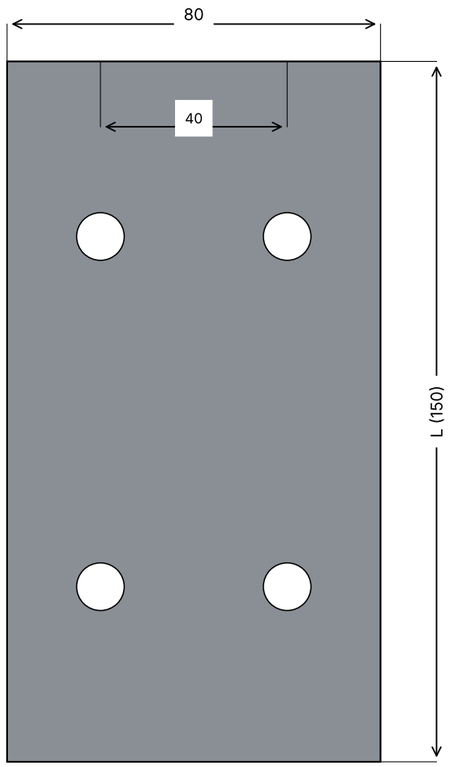
What the article does (RE-03 rev B, 2 Nov): assembled dry, the faces checked with a 0.05 feeler; cords in, bolts tightened from the middle out, squeeze read against torque; bead cured 24 h; end dams fitted; the space between the cords pressurised to 1 and 2 atm and logged (LC5, 7-14 Dec). Then pulled in prying through the lugs while the leak rate is read against the gap, which measures the gap at which the butyl stops sealing, the number the whole seal design turns on. Afterwards it stays assembled as the ageing sample.
Bolts: 2 + 2 at 75 pitch in the two 175 mm bays beside the 200 mm dummy fitting (RF-03), none inside it. Leg holes: 7 positions at 75 pitch (50 to 500), two rows, both legs.
Pull blocks: 160 × 80 × 550 (TP-04), 55 kg each in 4140; 19 kg each in 6061-T651 (DFM option). The seam model reaches the 0.11 mm gap at 3.75 kN per plain bolt; the pull on the article at that point comes from the D1-550 run, still owed. Fixture rated 50 kN.

HCSA TP-03 550 mm seal article with legs and the dummy hardpoint	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: both positions drawn; make the moved one, z 76 (agreed 10 Nov)	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked

Pull face (the face away from the joint), z measured from the frame's outer face

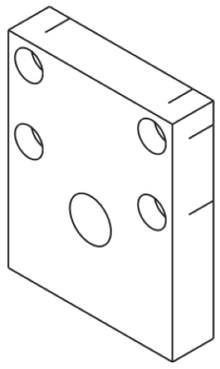


Bearing face (either), the leg screws' threads

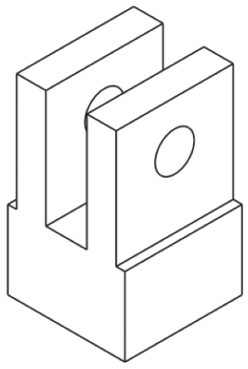


4 x M12 × 1.75 from each bearing face
(10.2 drill 31 deep, 26 full thread),
at the leg hole positions of TP-01.
Holes from the two faces line up;
160 - 2 × 31 leaves 98 mm of steel between.
The outer row sits 20 from the pull face.

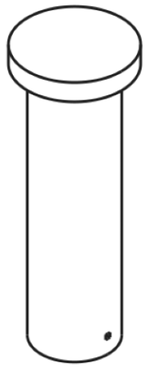
F-01: Lug plate 80 × 100 × 20, 20 H7 pin hole on the z 24 line, 4 x M12 × 45 counterbored



F-02: Fork: two 15 mm cheeks, 21 mm gap, 20 H8 holes, M24 × 3 female for the adapter



F-03: Pin 20 h6 × 70, 28 shoulder, E-clip groove, greased



Load train for run 2: crosshead, adapter, M24 stud, fork F-02, pin F-03, lug F-01, block, legs, frame, joint, and the mirror image below. Both pins parallel to the seam, so each end is free to turn. Checks (RE-03 rev B, 50 kN fixture load): pin double shear 377 kN; lug tear-out 720 kN; worst lug screw 39 kN against an 83 kN proof load. Material 4140 QT 28-32 HRC as drawn.

HCSA TP-04 Pull blocks (rev B interface) and the lug, fork and pin	
Test pieces, Rev F (180 frame), 8 Nov 2026. Replaces RE-01/02/03B, RF-03	mm. ISO 2768-m unless stated
Status: block 160 × 80 x L; M12 from both faces; M24 run 1; lug rows z 42/74; F-01..03 as 2 Nov	STEP files govern the geometry
Drawn: Claude for M. Theofanous Checked: - Released: -	For quotation. Not for machining until checked